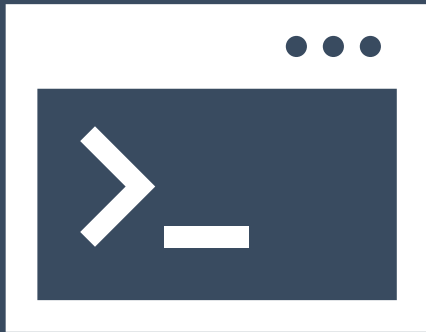


Introduction to Spatial Transcriptomics

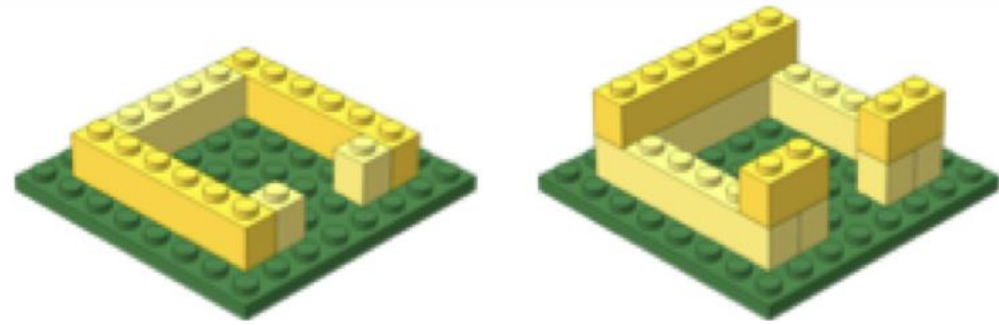
<https://tinyurl.com/hbc-spatial>



Harvard Chan Bioinformatics Core



Learning Objectives

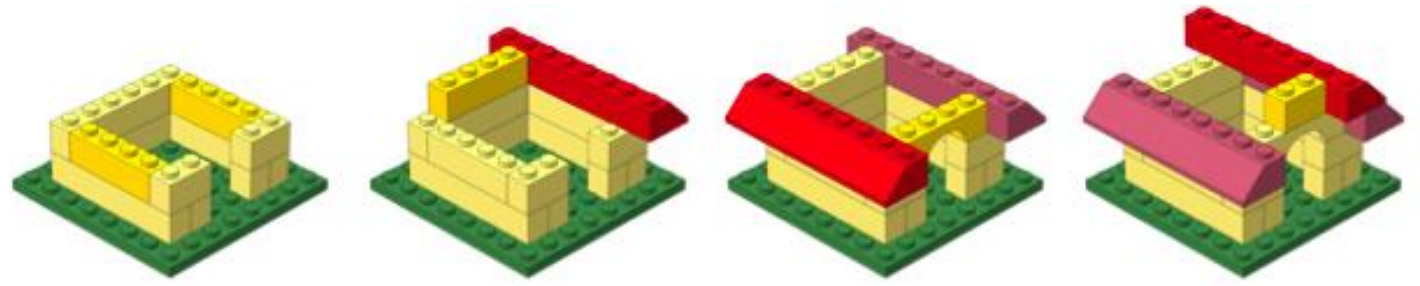


- Experimental design best practices of current spatial transcriptomics technologies
- Data loading and quality control of Visium HD samples
- Standard single-cell workflows adapted for spatial data, including highly variable gene selection, PCA, UMAP, kNN, and clustering
- Cell type annotation with deconvolution using RCTD for sequencing-based technologies
- Spatial-specific analyses that utilize the physical location of bins/cells on the tissue
 - Spatial clustering with BANKSY
 - Spatially variable gene detection with Moran's I
 - Cell-cell communication with CellChat

Exit survey

<https://tinyurl.com/hbc-spatial-survey>

Keep building!

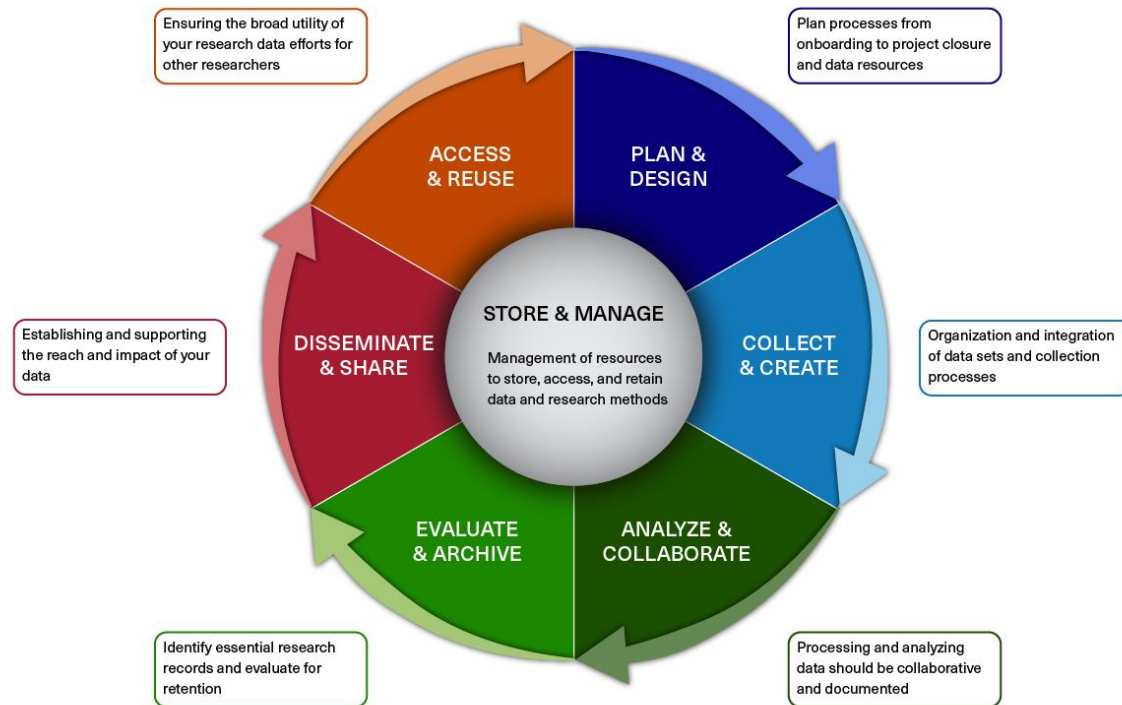


Module	Pre-requisite	Date	Time	Registration Page
Single Cell RNA-seq Analysis Using Cellenics	HarvardKey Credentials	10/21/26	9:30am-12:30pm	Register here
Multionics Part I*	Foundations in R	11/11/26	9:30am-12:30pm	TBD
Multionics Part II*	Multionics Part I	12/16/26	9:30am-12:30pm	TBD
Multionics Part III*	Multionics Part II	1/20/27	9:30am-12:30pm	TBD

<https://hsph.harvard.edu/research/bioinformatics/training/training-schedule/>

Research Data Management (RDM)

BIOMEDICAL RESEARCH DATA LIFECYCLE



Better RDM practice benefits you

- **HMS Data Management LMA**

- **Webpage:** <https://datamanagement.hms.harvard.edu>

- **Sign up** for quarterly email updates

- **Harvard-wide Research data Management**

- <https://researchdatamanagement.harvard.edu/>



Fall 2026 Data Lifecycle Training:

Learn the Lifecycle. Strengthen Your Research.

Plan & Design

September 22 & 24 
RDM: Concepts, Tools,
and Strategies

September 23 
RDM: Concepts, Tools,
and Strategies

September 30 
Onboarding and
Offboarding at Harvard

November 10 
Data Management and
Sharing Plans: From
Requirements to Reality

Collect & Analyze

September 24 
Introduction to
MATLAB

September 23 
The cBioPortal for
Cancer Genomics

October 15 
Introduction to
O2/Longwood

October 21 
Single Cell RNA-seq
Analysis Using Cellenics

December 3 
Introduction to
Python

Store & Evaluate

September 21 
Introduction to the General
Records Schedule

October 14 
Data Horror Stories:
Darkside of Data

October 20 
Offsite Storage at the
Harvard Depository

October 28 
Data Horror Stories:
From Ghoul to G.O.A.T.

December 1 
Managing Your Electronic
Records and Email

Share & Publish

November 18 
Federal Public Access
Policies

December 2 
Research Data Sharing:
Principles, Practices, and
Platforms

 In-person workshop
 Virtual Zoom training

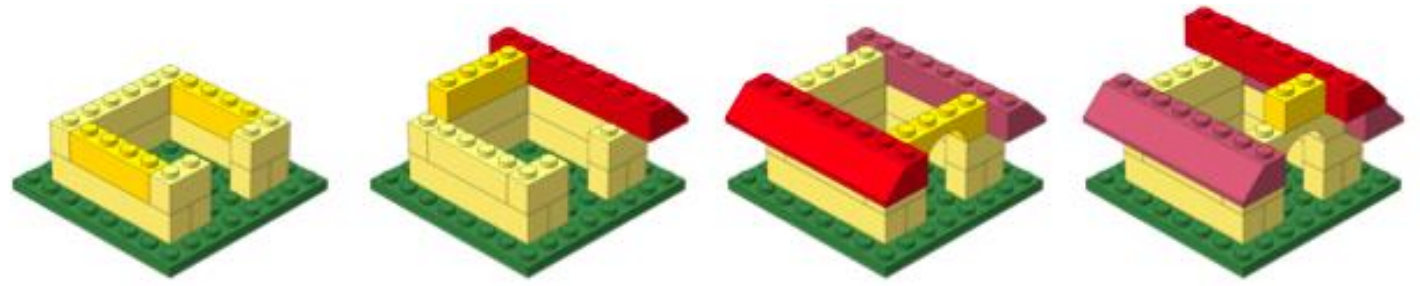


Learn More & Register
bit.ly/rdmwg-calendar



<https://datamanagement.hms.harvard.edu/training-events/rdmwg-calendar>

Keep building!



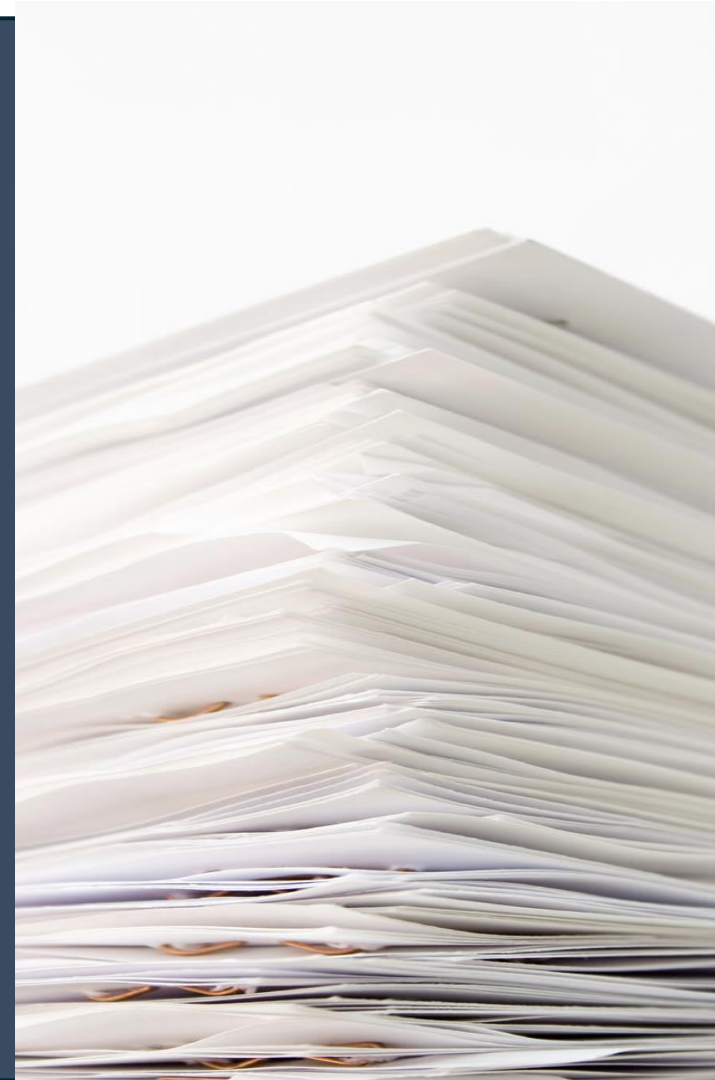
Workshop	Dates	Time	Location	Registration Page*
Introduction to Python	November 10, 13, 17 & 20, 2026	10am-12pm	Zoom	Register here
Introduction to R	May 4, 7, 11 & 14, 2027	9:30am-12pm	Zoom	Register here

Workshop	Pre-requisite*	Dates	Time	Location	Registration Page**
Tools for Reproducible Research	Introduction to R	October 27, 30 & November 3, 2026	9:30am-12pm	Zoom	Register here
Introduction to Spatial Transcriptomics	Introduction to R	January 26, 29, February 2 & 5, 2027	10am-12pm	Zoom	TBD
Introduction to scRNA-seq (R-based)	Introduction to R	February 16, 19, 23 & 26, 2027	10am-12pm	Zoom	Register here
Pseudobulk and related approaches for scRNA-seq analysis	Introduction to R	March 16, 19, 23 & 26, 2027	10am-12pm	Zoom	Register here

<https://hsph.harvard.edu/research/bioinformatics/training/training-schedule/>

Talk to us early!

Involvement in study design to optimize experiments



More Information

- *HBC training materials: <https://hbctraining.github.io/main>*
- *HBC website: <https://hsph.harvard.edu/research/bioinformatics/>*

Contact Us

Sign up for our mailing list:

<https://tinyurl.com/hbc-training-mailing-list>

- *HBC training team:* hbctraining@hsph.harvard.edu
- *HBC consulting:* bioinformatics@hsph.harvard.edu